

CBSE Class 10 Mathematics — Surface Areas and Volumes

Time Allowed: 3 Hours

Maximum Marks: 80

General Instructions:

1. This question paper contains *questions.Allquestionsarecompulsory.*
1. Questions are grouped into sections A through E.
2. Use of calculators is NOT permitted.

Section A: Multiple Choice & Assertion-Reason (1 mark each)

1. If the radius of a sphere is 7 cm, then its surface area is:
(A) (A) 308 cm^2
(B) (B) 440 cm^2
(C) (C) 616 cm^2
(D) (D) 1232 cm^2
2. The volume of a right circular cylinder with radius 3 cm and height 7 cm is:
(A) (A) 154 cm^3
(B) (B) 198 cm^3
(C) (C) 264 cm^3
(D) (D) 396 cm^3
3. The total surface area of a cube whose edge is 5 cm is:
(A) (A) 150 cm^2
(B) (B) 125 cm^2
(C) (C) 100 cm^2
(D) (D) 250 cm^2
4. If the slant height of a cone is 10 cm and the radius is 6 cm, its curved surface area is:
(A) (A) $30\pi \text{ cm}^2$
(B) (B) $40\pi \text{ cm}^2$
(C) (C) $50\pi \text{ cm}^2$
(D) (D) $60\pi \text{ cm}^2$
5. The lateral surface area of a cuboid of dimensions $8 \text{ cm} \times 6 \text{ cm} \times 5 \text{ cm}$ is:

- (A) (A) 70 cm^2
 (B) (B) 110 cm^2
 (C) (C) 140 cm^2
 (D) (D) 280 cm^2
6. Assertion (A): The total surface area of a solid hemisphere of radius r is $3\pi r^2$. Reason (R): The total surface area of a sphere is $4\pi r^2$.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 (C) (C) Assertion (A) is true but Reason (R) is false
 (D) (D) Assertion (A) is false but Reason (R) is true
7. Assertion (A): If the radius of a cylinder is doubled and its height is halved, the volume remains the same. Reason (R): The volume of a cylinder is given by $V = \pi r^2 h$.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 (C) (C) Assertion (A) is true but Reason (R) is false
 (D) (D) Assertion (A) is false but Reason (R) is true
8. Assertion (A): The volume of a cone with radius r and height h is $\frac{1}{3}\pi r^2 h$. Reason (R): A cone is one-third of a cylinder with the same base and height.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 (C) (C) Assertion (A) is true but Reason (R) is false
 (D) (D) Assertion (A) is false but Reason (R) is true
9. Assertion (A): The slant height l of a cone is given by $l = \sqrt{h^2 + r^2}$. Reason (R): This is derived from Pythagoras theorem applied to the triangle formed by height, radius, and slant height.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 (C) (C) Assertion (A) is true but Reason (R) is false
 (D) (D) Assertion (A) is false but Reason (R) is true

10. Assertion (A): If the surface area of a sphere is 616 cm^2 , its radius is 7 cm. Reason (R): The surface area of a sphere is $4\pi r^2$.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true

Section C: Short Answer I (3 marks each)

1. Two cubes each of volume 64 cm^3 are joined end to end. Find the surface area of the resulting cuboid.
2. A metallic sphere of radius 4.2 cm is melted and recast into the shape of a cylinder of radius 6 cm. Find the height of the cylinder.
3. The slant height of a frustum of a cone is 4 cm and the perimeters of its circular ends are 18 cm and 6 cm. Find the curved surface area of the frustum.
4. A solid is in the form of a right circular cone mounted on a hemisphere. If the radius of the hemisphere and the cone is 3.5 cm and the height of the cone is 4 cm, find the volume of the solid. (Use $\pi = 22/7$)
5. The diameter of a metallic ball is 4.2 cm. What is the mass of the ball, if the density of the metal is 8.9 g/cm^3 ?

Section D: Long Answer (4-5 marks each)

1. A metallic sphere of radius 6 cm is melted and recast into the shape of a cylinder of radius 4 cm. Find the height of the cylinder.
2. Two cubes each of volume 64 cm^3 are joined end to end. Find the surface area of the resulting cuboid.
3. A conical tent is 10 m high and the radius of its base is 24 m. Find the slant height of the tent and the cost of the canvas required to make the tent, if the cost of 1 m^2 canvas is Rs. 70.
4. A medicine capsule is in the shape of a cylinder with two hemispheres stuck to each of its ends. The length of the entire capsule is 14 mm and the diameter of the capsule is 5 mm. Find its surface area.
5. A hemispherical tank, full of water, is emptied by a pipe at the rate of $3\frac{4}{7}$ litres per second. How much time will it take to empty half the tank, if it is 3 m in diameter? (Take $\pi = \frac{22}{7}$)

6. A solid toy is in the form of a hemisphere surmounted by a right circular cone. The height of the cone is 2 cm and the diameter of the base is 4 cm. If a right circular cylinder circumscribes the toy, find the difference between the volume of the cylinder and the volume of the toy. (Take $\pi = 3.14$)
7. A metallic sphere of radius 6 cm is melted and recast into the shape of a right circular cylinder of radius 6 cm. Find the height of the cylinder.
8. A bucket is in the form of a frustum of a cone with a capacity of 12308.8 cm^3 . The radii of the top and bottom circular ends are 20 cm and 12 cm respectively. Find the height of the bucket. (Use $\pi = 3.14$)
9. A medicine capsule is in the shape of a cylinder with two hemispheres stuck to each of its ends. The length of the entire capsule is 14 mm and the diameter of the capsule is 5 mm. Find its surface area.